provide the first evidence for dimensions beyond those directly visible, taking us one step closer to establishing superstring theory as the long-sought unified theory.

If superstring theory is proven correct, we will be forced to accept that the reality we have known is but a delicate chiffon draped over a thick and richly textured cosmic fabric. Camus' declaration notwithstanding, determining the number of space dimensions—and, in particular, finding that there aren't just three—would provide far more than a scientifically interesting but ultimately inconsequential detail. The discovery of extra dimensions would show that the entirety of human experience had left us completely unaware of a basic and essential aspect of the universe. It would forcefully argue that even those features of the cosmos that we have thought to be readily accessible to human senses need not be.

Past and Future Reality

With the development of superstring theory, researchers are optimistic that we finally have a framework that will not break down under any conditions, no matter how extreme, allowing us one day to peer back with our equations and learn what things were like at the very moment when the universe as we know it got started. To date, no one has gained sufficient dexterity with the theory to apply it unequivocally to the big bang, but understanding cosmology according to superstring theory has become one of the highest priorities of current research. Over the past few years, vigorous worldwide research programs in superstring cosmology have yielded novel cosmological frameworks (Chapter 13), suggested new ways to test superstring theory using astrophysical observations (Chapter 14), and provided some of the first insights into the role the theory may play in explaining time's arrow.

The arrow of time, through the defining role it plays in everyday life and its intimate link with the origin of the universe, lies at a singular threshold between the reality we experience and the more refined reality cutting-edge science seeks to uncover. As such, the question of time's arrow provides a common thread that runs through many of the developments we'll discuss, and it will surface repeatedly in the chapters that follow. This is fitting. Of the many factors that shape the lives we